11.6 Updated National Immunization Schedule Chart

Immunization/vaccination produce a response in the body that is similar to the body's response in the body that is similar to the body's response to a natural infection (Table 11.4). Immunization or vaccines can therefore protect the body from a disease before the disease has a chance to cause illness. Immunization has helped to reduce the impact of communicable disease on health and well being. Some diseases have been well controlled and other has been eliminated from some parts of the world because of vaccination. Stopping vaccination may lead to epidemic.

Summary

The reaction between antigen and antibody is highly specific. It is compared to the lock and key system. **ELISA** (Enzyme-Linked Immuno Sorbent Assay) is a plate-based assay technique designed for detecting and quantifying substances such as peptides, proteins, antibodies and hormones. There are **four kinds** of ELISA assay tests. They are: Direct ELISA, Indirect ELISA, Sandwich ELISA and Competitive ELISA. Western blotting technique is used for the identification of particular protein from the mixture of proteins. The most common protein sample used for Western blotting is cell lysate. Blotting refers to the transfer of the protein from the gel to the nitrocellulose paper by capillary action. The substances causing allergic/hypersensitivity is known as allergens. Allergic rhinitis develops when the body's immune system becomes sensitized and overreacts to something in the environment like pollen grains, strong odour of perfumes, dust etc. Certain drugs such as penicillin, cephalosporin and streptomycin can absorb non-specifically to protein on surface of RBC forming complex similar to hapten-carrier complex.

Transfer of living cells, tissues or organs from one part of the body to another or from one individual to another is known as transplantation. The graft tissue antigens induce an immune response in the host. This type of immune response is called host versus graft reaction. The ultimate goal of any immunization program is the eradication of the disease. Active natural immunization involves activation of immune system in the body to produce antibodies. It is achieved in both clinical and subclinical infections Immunization has helped to reduce the impact of communicable disease on health and well being.

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